

Code in Place

2026

Stanford CS106A

Section - Week 1

Welcome to Section!

David Tsai, Code in Place

Today's Agenda



Introductions

Icebreakers



Norms

Guidelines



Concept Review

Karel, Control Flow



Practice Problems

"Hospital Karel"

Please Turn On Your Camera



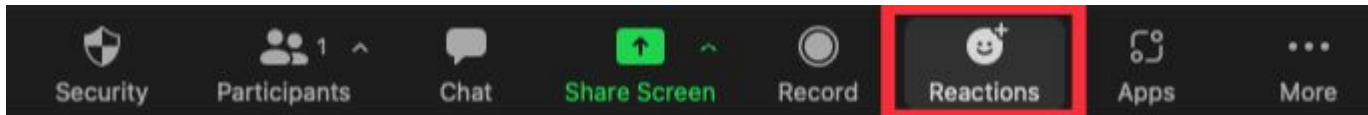
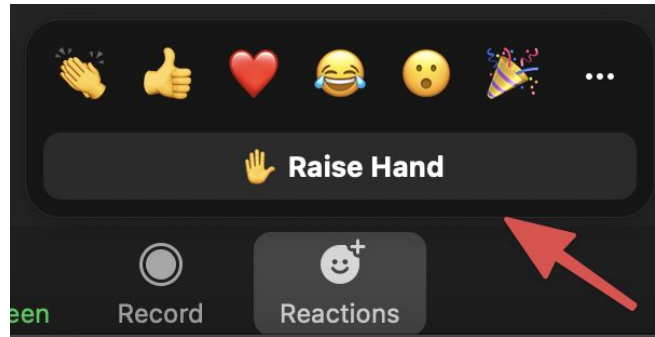
If you're able, please **turn on your camera!**
.... It can really make the section come to life!



(Image source: <https://as.virginia.edu/eight-ways-get-more-out-zoom>)

Zoom Reactions

- 👍 **Thumbs Up:** If you understand.
- 🙋 **Raise Hand:** If you have a question (or just speak in the mic).



Introductions

Hi, I'm David!

- Head TA, here at my 5th Code in Place.
 - Started as a CIP student! 2x volunteer Section Leader.
- CS @ Massachusetts Institute of Technology (MIT)
- Product Manager and former Software Engineer
- Love photography, video games, movies
- Guilty pleasure: Reality competition shows like Survivor
- Cool superpower I'd like to have:
 - Felix Felicis ("Liquid Luck" potion) from Harry Potter

How about you?

- Where are you from?
- Your coding experience
- Cool superpower you'd like to have

Course Weekly Timeline

Lectures

2 Lessons

Both released every Monday.
~1 hour per lecture.

Section

Live Zoom Session

Practice lecture concepts.
Prepare you for assignment.

Assignment

Coding Challenges

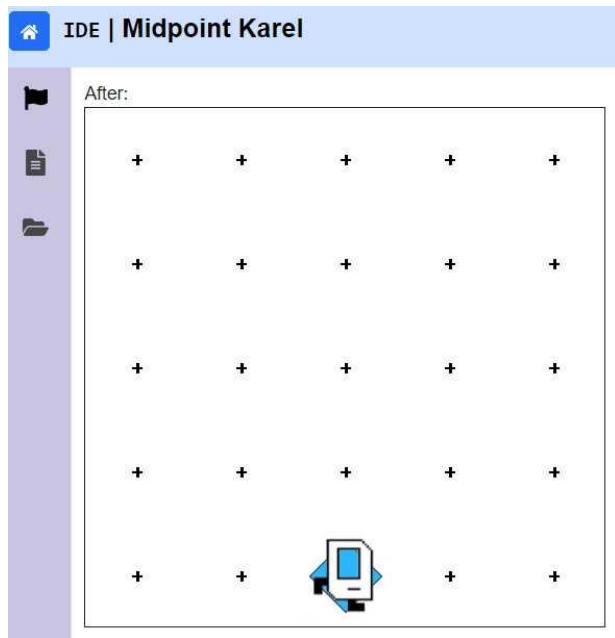
Solidify your understanding.
Optional: "Extension" projects

Brief Tour of Course's Web Platform

The screenshot displays the Code in Place web platform interface. The browser address bar shows the URL `codeinplace.stanford.edu/cip3/join/ide/a/piles`. The interface is divided into several sections:

- IDE | Piles**: The main header with a **Run** button and a **Share** button.
- Instructions**: A sidebar on the left containing a **Restart** button and a task description: "Your task: Write a program in the editor, that makes Karel pick up all the beepers on the first row of this world." Below this, it says "After your program finishes, Karel's world should have no beepers, like so:" followed by a 6x6 grid of beepers. A "Text descriptions" toggle is visible.
- main.py**: The central code editor showing a Python program for Karel. The code includes imports, file names, and functions for moving, picking up beepers, and a main function that loops through the first row of the world.
- World**: A sidebar on the right showing a 6x6 grid of beepers. Below the grid, there are "Text descriptions" and a slider control.
- Terminal**: A black bar at the bottom of the interface.

Some Basic Python Functionality is TURNED OFF....for now!

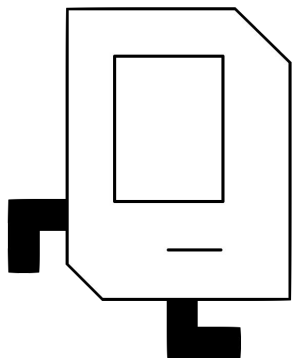


For the duration of Karel unit (Weeks 1 & 2):

- Basic Python functionality is turned off for simplicity
 - Focus on fundamentals of programming
- Examples of disabled functionality:
 - print statements
 - variables, lists, dictionaries
 - conditional keywords (**NOT**, etc.)
 - function return statements

Concepts Review (Lightning Round!)

Karel is like a LEGO set with only 4 types of blocks....but you can do a lot!



Karel only knows four commands at the beginning:

`move()`

`turn_left()`

`put_beeper()`

`pick_beeper()`

Helpful website to visualize Karel (by Sophia Westwood):

www.karelhelper.com

Control Flow



for **Loops**



while **Loops**



if/else

for Loop

```
for i in range(count):  
    statements                # note indenting
```

```
# Example:  
def turn_right():  
    for i in range(3):  
        turn_left()           # note indenting
```

while Loop

```
while condition:  
    statements                # note indenting
```

```
# Example:  
def move_to_wall():  
    while front_is_clear():  
        move()                # note indenting
```



Conditions Karel Can Check For

<i>Test</i>	<i>Opposite</i>	<i>What it checks</i>
<code>front_is_clear()</code>	<code>front_is_blocked()</code>	Is there a wall in front of Karel?
<code>left_is_clear()</code>	<code>left_is_blocked()</code>	Is there a wall to Karel's left?
<code>right_is_clear()</code>	<code>right_is_blocked()</code>	Is there a wall to Karel's right?
<code>beepers_present()</code>	<code>no_beepers_present()</code>	Are there beepers on this corner?
<code>beepers_in_bag()</code>	<code>no_beepers_in_bag()</code>	Any there beepers in Karel's bag?
<code>facing_north()</code>	<code>not_facing_north()</code>	Is Karel facing north?
<code>facing_east()</code>	<code>not_facing_east()</code>	Is Karel facing east?
<code>facing_south()</code>	<code>not_facing_south()</code>	Is Karel facing south?
<code>facing_west()</code>	<code>not_facing_west()</code>	Is Karel facing west?

Check out “Karel Reader”, Chapter 10 for full reference

Which Type of Loop Should You Use?

for Loop

You know exactly how many times to repeat (definite loop)

vs.

while Loop

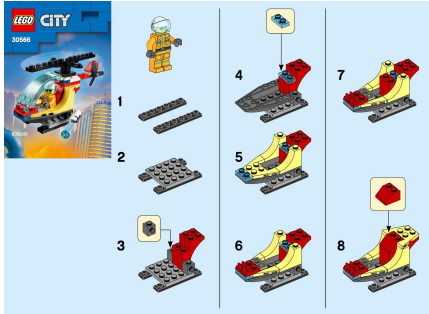
You don't know exactly how many times to repeat (indefinite loop)

if-else Statement

```
if condition:  
    statements                # note indenting  
else:  
    statements
```

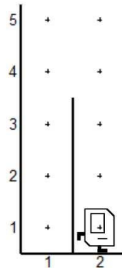
```
# Example:  
def invert_beepers():  
    if beepers_present():  
        pick_beeper()        # note indenting  
    else:  
        put_beeper()         # note indenting
```


Decomposition



Break down a problem into more manageable sub-problems

- “Divide and conquer” approach
- Which sorts of tasks should be decomposed?
- Rule of Thumb: Each function should execute a single purpose.
- Code becomes easier to read! For both you and everyone else.



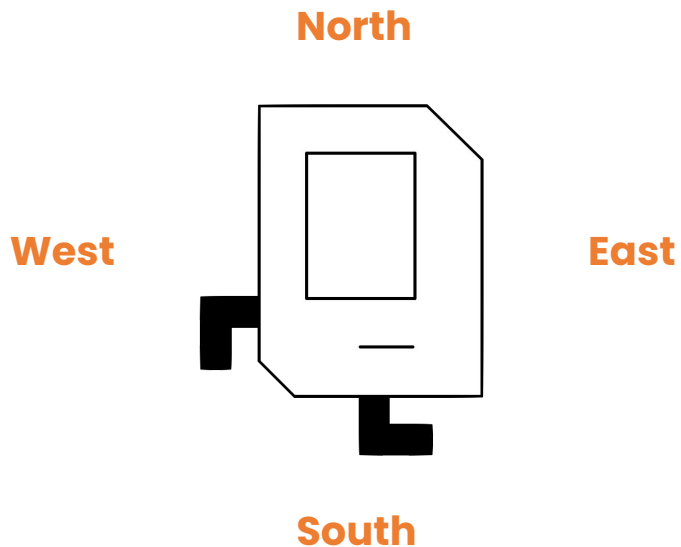
```
turn_left()
while right_is_blocked():
    move()
turn_right()
move()
turn_right()
move_to_wall()
turn_left()
```



```
ascend_hurdle()
move()
descend_hurdle()
```

Navigating Karel: Compass Direction vs. Relative Direction

Compass Direction



Navigating Karel: Compass Direction vs. Relative Direction

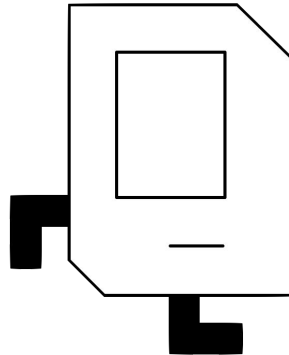
Compass Direction

VS.

Relative Direction

North

West



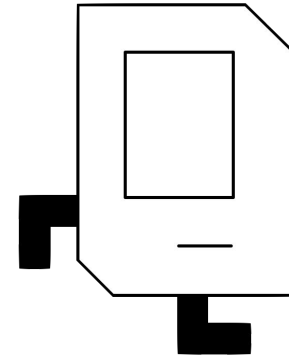
East

South

```
def face_west():  
    while not_facing_west():  
        turn_left()
```

Left

Back

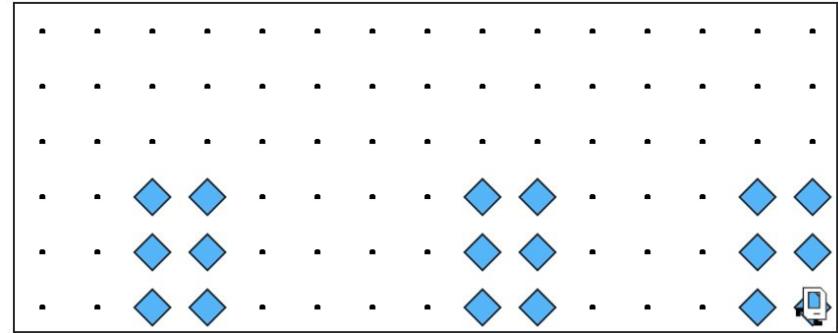
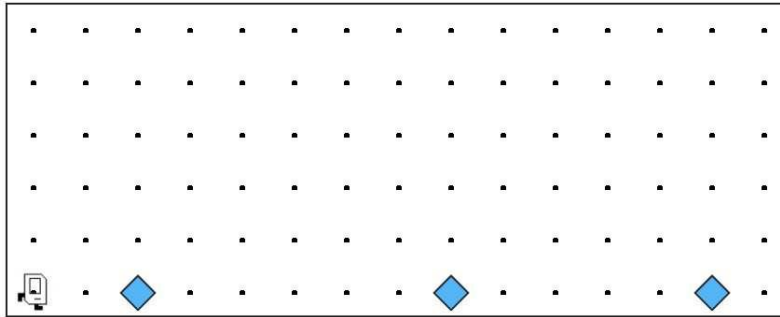


Front

Right

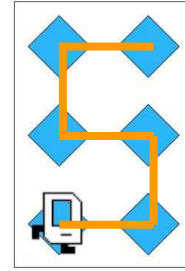
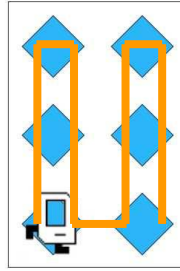
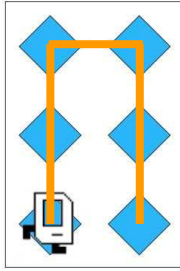
```
def turn_around():  
    for i in range(2):  
        turn_left()
```

Section Exercise: “Hospital Karel”



Section Exercise: "Hospital Karel"

Which option would you pick to build the 2x3 hospital? Pros vs Cons?



What if the hospital was another shape?

